Summary of activities, 23 March 26

1. Reprogramming PD07 fibroblasts to iPSCs (continued)

I have continued Days 5–16 of my PD07 reprogramming by feeding the cells with fresh NutriStem everyday. There is a clear change toward a more stem cell–like morphology and we are beginning to see the emergence of likely colonies (see Fig. 1). I will continue to care for the cells, and Bin and I are monitoring for the appropriate time to pick clones (likely after Day 14, based on Bin's previous experiences).

On Friday (27Mar26), Bin, Kritika, and I met with the team from Portal (Darby and Alix) to discuss the other reprogramming conditions we want to test (linear vs. circular and modified vs. non-modified mRNA; lipofectamine vs. mechanoporation). We have made a plan to begin testing these in April, contingent on the arrival of several factors we need. To ensure we are prepared, Kritika and I did an accounting of the stocks and aliquots in the boxes labeled `RNA kit` and `miRNA` in the −80°C. I sent these to Bin for her review so that we can order all necessary materials.

2. GFP electroporation of PD05 fibroblasts

On Tuesday (24Mar26), Bin, Kritika, and I finalized the worksheet I made for GFP electroporation. I also made these edits to my general worksheet for fibroblast electroporation. On Wednesday (25Mar26), Kritika and I electroporated her PD05 fibroblasts with pmaxGFP; I operated and Kritika scribed. Things went smoothly, with the exception of inaccurate cell counts, and Kritika and I had achieved GFP expression by the next day (see Fig. 2). This is detailed on the worksheet in my LabArchives.

Regarding the cell counter, subsequent investigation with Bin also revealed that cell count data was apparently not being consistently saved on the Luna FX7. I have added monitoring the Luna FX7 memory and periodically exporting its data cache (to Teams) to my TC maintenance worksheet (see Section 4) to ensure this error does not recur. We met with Tatyana on Friday (27Mar26) and made two observations: (1) the chamber must be loaded from the correct loading port on

the right; and (2) inserting the slide fully into the port ensures no band artifact, no elevated background noise, and more accurate counts. I suspect this was the source of our inaccurate counts on Wednesday.

3. T32 application

I began preparing application materials for the MGH T32 program in Precision & Genomic Medicine. I made a list of candidate faculty mentors to sponsor my application and worked with Bin and Dr. Schweitzer to draft an email inquiry. I will continue to prepare the other primary materials (personal and research statements) over the coming week to ensure Bin and Dr. Schweitzer has sufficient time to proofread my drafts and provide comments.

4. TC incubator sterilization and other maintenance

On Monday (23Mar26), Tatyana showed me how to steri-cycle the incubator. I completed sterilization of the top incubator on Tuesday (24Mar26), and we also auto-started the bottom incubator on Thursday (26Mar26). I have noted these dates in my weekly TC maintenance worksheet so that I can ensure incubator care remains on a consistent schedule. I also completed the other relevant TC duties; the worksheet is attached. I continued to assist Kritika in cataloguing our TC reagent stocks and aliquots (updated in the inventory on Teams), and we set aside explicit shelves in the Open Lab and TC for reagents.

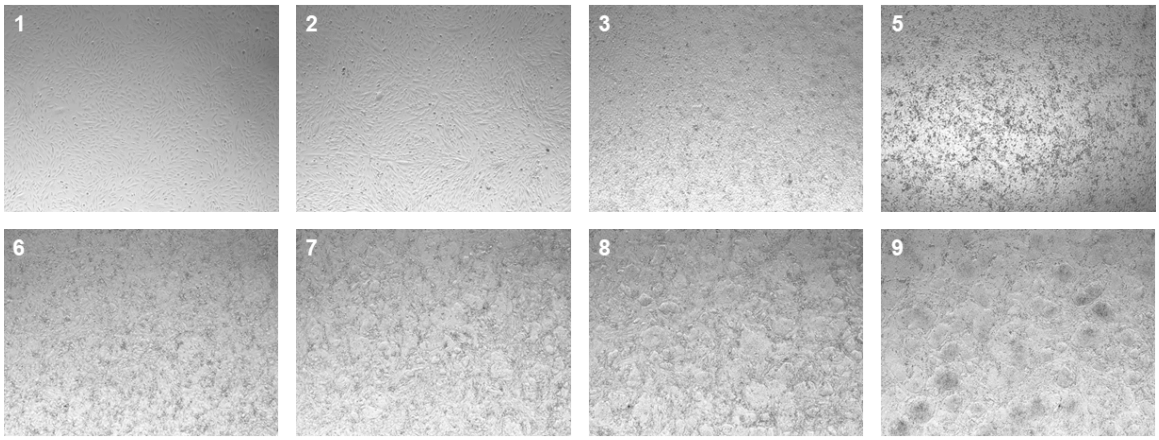

Fig. 1. Phase microscopy (4X) timecourse of reprogramming PD07 fibroblasts to hiPSCs using the StemGent Stem RNA 3rd Gen Kit, days 1–9 (as indicated).

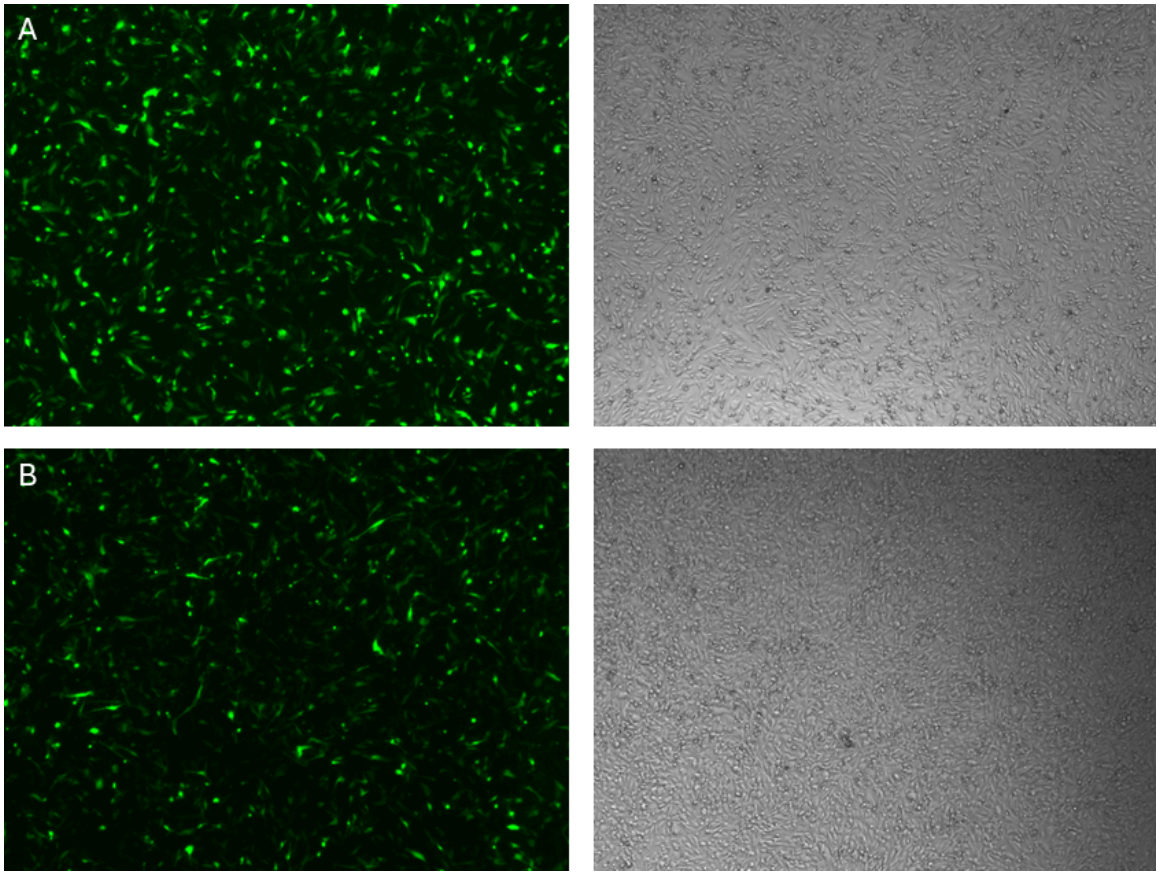

Fig. 2. Successful electroporation of two wells (A, B) of PD05 fibroblasts. Left: phase; right: GFP.

MWKST-TC-maintenance-HH

Massachusetts General Hospital Founding Member, Mass General Brigham	George A. "Doc" Lopez, MD Laboratory for Regenerative Cell Therapy White 536, 55 Fruit St, Boston, MA 02114

Process	TC equipment maintenance		
Date & time	25Mar26	Operator	HHsing

Tasks:

Instructions	Data	Scribe
Prepare fresh 70% isopropyl alcohol	ø	HH
Clean:	ø	HH
Surfaces	ø	HH
Shelves	ø	HH
Doors	ø	HH
Chairs	ø	HH
Cart (including wheels)	ø	HH
Plastic tubs (beneath vacuums)	ø	HH
Floor	ø swept	HH
Take out the trash:	ø	HH
Regular	ø	HH
Bio-hazard	ø	HH
Replace sharps containers: spent sharps containers go in the bio-hazard trash; fresh sharps containers can be found beneath the sink	~~ø~~ N/A Number remaining? 2 Request more? N/A	Still empty HH
Empty vacuums	~~ø~~ N/A	Not full HH
Replace consumables	ø	HH
Charge pipettors	ø	HH
Incubator:	CO_2: 5; 5 Temp.: 37; 37	HH
Last sterilization?	Top: 24Mar Bottom: ——	HH
Replace water	ø 25Mar26	HH
Maintenance?	Needed: □ N/A: ø	HH
Fridge/freezer maintenance?	Needed: □ N/A: ø	HH
BSC recertification?	Needed: □ N/A: ø	HH
Microscope maintenance?	Needed: □ N/A: ø Memory: 5 GB	HH
Centrifuge maintenance?	Needed: □ N/A: ø	HH
Pipette calibration?	Needed: □ N/A: ø	HH
Dewar: replace liquid nitrogen?	Needed: □ N/A: ø	HH

All steps completed? ø

Scribe: HH / 27Mar26
(Initial & date)

Luna FX7 available memory: 202.8 GB; data exported to Teams

Notes: Ordered more 50-mL Luer-lock syringes (1 box remaining) 25Mar26

2L 70% IPA 26Mar26 — order more 99% IPA? Only 2 non-GMP containers left

Empty cell counter to USB weekly?